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**CPC452 Animation & Virtual Reality
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Assignment 2 Report

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1 Abstract

Toxland VR: Hop & Help is an educational Virtual Reality application that teaches young children to recognise and safely handle toxic or hazardous household materials such as medicines, household poisons, pesticides, and aerosol sprays. Inspired by the classic Snakes and Ladders board game, players roll a virtual dice to move across an interactive board and answer *Betul/Salah* (True/False) questions about safe household practices: correct answers earn points and a short explanation, wrong answers deduct points, while ladder and trap squares on the board provide the familiar advance/fall-back movement. The educational content is presented in Bahasa Melayu. This project converts and adapts an existing Unity3D VR build developed with the Meta SDK for the Meta Quest headset into an XR-compatible build that runs on the PICO 4 headset using the PICO XR SDK. The conversion covers Unity XR settings, controller input mapping, camera rig and locomotion, UI interaction, and Android build configuration. This report documents the VR concept, background study, conversion methodology, system development, and testing evidence confirming that the application runs successfully on the target platform.

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2 VR Concept

This section presents the conceptual design of Toxland VR: Hop & Help, including its theme, intended audience, world design, storyline, storyboard, and team structure.

2.1 Concept Theme

The concept theme of Toxland VR is household hazard safety education, delivered through a board-game-inspired progression system. The application is a **hybrid of an educational and entertainment-focused experience**: it borrows the familiar mechanics of Snakes and Ladders — ladders that let players climb ahead and snakes that send them back. Rather than every tile carrying the same activity, the board spaces come in several kinds: *regular* spaces that simply pass the turn along; *Betul/Salah* (True/False) question spaces that pose a safety statement and reward correct knowledge while correcting misconceptions; **red boxes**, which act as ladders — they present a *good* toxicology-related behaviour and, once acknowledged, move the player forward; and **blue boxes**, which act as snakes — they present a *poor* or unsafe behaviour and move the player back. Together these reinforce real-world household safety habits around medicines, household poisons, pesticides, aerosol sprays, and supplements. The educational intent is primary (the learning outcome is the goal), while the game framing, scoring, and rewards exist to sustain motivation and engagement. The design objectives are to help children (i) recognise potentially hazardous household items, (ii) adopt safe storage practices, (iii) understand warning labels and safety symbols, (iv) behave responsibly around household chemicals, and (v) retain these lessons through interactive, positively reinforced repetition.

2.2 Target Audience

The primary target audience is **young children, approximately 6–11 years old (lower- to upper-primary school)**. This range is chosen for three reasons. First, it is the developmental window in which children begin to access kitchens, bathrooms, and storage areas with reduced adult supervision, making household-chemical awareness directly relevant to their daily lives. Second, children of this age respond strongly to game-based reward/penalty structures, which the Snakes-and-Ladders mechanic exploits

directly. Third, because reading ability varies widely across this band, the interface deliberately pairs short Bahasa Melayu text with icons, colour coding, and audio cues so that the safety content remains accessible to weaker readers. The VR medium is selected because immersive, first-person decision-making is expected to reinforce safety habits more effectively than passive instruction such as posters or textbook lessons, while the playful framing keeps the experience age-appropriate rather than frightening.

2.3 World Design

Players enter the colourful world of Toxland, where everyday household hazards are turned into a board-game challenge. The level is a single numbered board laid out as a winding path of tiles in the Snakes-and-Ladders tradition. The tiles are of several kinds: plain pass-through spaces, *Betul/Salah* question spaces, red boxes that act as ladders (climb ahead), and blue boxes that act as snakes (fall back). Rather than the board being physically divided into rooms, the safety content placed on its tiles covers the hazards found throughout a typical home:

- **Medicines** — reading labels (e.g. red labels marking external-use medicines), safe dosage, and storage out of children’s reach.
- **Household poisons** — rat poison and insecticides, and where they may (or may not) be kept.
- **Pesticides and aerosols** — safe storage away from heat and open flame.
- **Supplements** — vitamins and iron, and the danger of taking too much at once.
- **Poisoning response** — what should (and should not) be done if poisoning occurs.

Question tiles present a *Betul/Salah* (True/False) statement with immediate Right/Wrong feedback and a short explanation, while red and blue box tiles present a good or unsafe behaviour and move the player forward or back accordingly. The visual style is bright and cartoon-like to remain non-threatening for young children. The board layout is shown in the level design plan in Section 4.5.

2.4 Storyline

Players travel across the Toxland board to reach the final space. On each turn they:

1. Roll a virtual dice and move along the board to the next space.
2. Land on a space, whose effect depends on its type:
 - A *Betul/Salah* question space presents a household-safety statement; the player decides *Betul* (true) or *Salah* (false) with the VR controller and receives immediate Right/Wrong feedback with a short explanation, gaining (+2) or losing (−2) points accordingly.
 - A **red box** (ladder) presents a good toxicology-related behaviour and, once acknowledged, moves the player forward.
 - A **blue box** (snake) presents a poor or unsafe behaviour and moves the player back.
 - A regular space simply passes the turn along.

The game ends when the player reaches the final space, at which point the final score is tallied and displayed on the end screen.

2.5 Storyboard

Figure 1 presents the six-frame storyboard for the core gameplay loop.

<i>[Insert screenshot]</i> <i>Frame 1 — Welcome screen introducing the game.</i> <code>assets/storyboard-1.png</code>	<i>[Insert screenshot]</i> <i>Frame 2 — Entering the Toxland board and rolling the dice.</i> <code>assets/storyboard-2.png</code>
<i>[Insert screenshot]</i> <i>Frame 3 — Landing on a red box (good toxicology behaviour) → a ladder that moves the player forward.</i> <code>assets/storyboard-3.png</code>	<i>[Insert screenshot]</i> <i>Frame 4 — Landing on a blue box (unsafe behaviour) → a snake that sends the player back.</i> <code>assets/storyboard-4.png</code>
<i>[Insert screenshot]</i> <i>Frame 5 — A Betul/Salah safety question with its explanation.</i> <code>assets/storyboard-5.png</code>	<i>[Insert screenshot]</i> <i>Frame 6 — Reaching the final space and the end screen with the final score.</i> <code>assets/storyboard-6.png</code>

Figure 1: Six-frame gameplay storyboard for Toxland VR: Hop & Help.

2.6 Team Structure

Table 1: Project team structure and responsibilities

Name	Role	Responsibility
Lai Yicheng	Project Director	Project planning, scheduling, and management; coordination and report compilation
Koay Chun Keat	Programmer	Unity scripting, game logic, scoring system, and Meta-to-PICO migration
Yong Pei Yi, Angeline	3D Artist	Board design, household objects, and environment modelling
Tan Ke Jun	Animator	Character and object animations
Tan Yi Pin	UI/UX Designer	Menus, educational interfaces, and feedback systems

3 Background Study

3.1 Existing Applications

To position Toxland VR against the state of the art, we reviewed three existing, peer-reviewed serious-game and VR applications for child safety and hazard education. Table 2 compares them on purpose, platform, and interaction model.

Table 2: Comparison with existing child-safety / hazard-awareness applications

Application	Purpose	Platform	Interaction Model
Safe City [1]	Mobile serious game teaching home, road, and sports safety knowledge and behaviour to primary-school children (grades 4–6)	Mobile phone (Android/iOS)	Avatar-based role-play; touch-screen taps through scenario mini-games
FSCHOOL Fire Preparedness [2]	VR serious game for fire-preparedness training in the school context	Immersive VR (CAVE virtual environment)	First-person immersive navigation and interaction with simulated fire scenarios
Adaptive Road-Safety Game [3]	Adaptive game-based learning for children’s road-safety education and practice	VR built in Unity with a Kinect sensor	3D first-person play with body-gesture / motion sensing and difficulty that adapts to the learner
Toxland VR (this work)	Household toxic-material handling, storage, and label awareness for young children	Standalone VR (PICO 4, Android)	Snakes-and-Ladders board progression; controller ray selection, grab, dice roll, teleport, and joystick locomotion

Educational objectives. All three reviewed systems target child safety, but each addresses a different hazard domain: Safe City covers general home/road/sports safety [1], FSCHOOL focuses on fire preparedness [2], and Khan et al. address road safety [3]. None specifically target *household toxic and chemical materials*, which is the niche Toxland VR

fills.

Interaction methods. Safe City is the least immersive — it is a flat, touch-driven mobile game [1]. The fire-preparedness game uses a CAVE projection environment that surrounds the learner but is fixed to specialised hardware [2]. The road-safety system adds embodied, gesture-based input via Kinect and, notably, *adaptive difficulty* [3]. Toxland VR uses a consumer head-mounted display (PICO 4) with six-degrees-of-freedom controllers, giving room-scale immersion at a far lower cost and footprint than a CAVE.

User engagement and learning effectiveness. A common thread across these works is that game framing — avatars, missions, points, and reward/penalty feedback — sustains children’s engagement and supports knowledge retention better than passive instruction [1, 3]. Toxland VR operationalises exactly this principle: each question tile delivers immediate Right/Wrong feedback paired with a short explanation, so the safety lesson is coupled directly to the player’s own answer.

3.2 Key Innovation

Toxland VR builds on the reviewed applications in three distinct ways:

1. **A board-game progression mechanic.** Rather than free-roaming scenarios, Toxland VR structures learning as a Snakes-and-Ladders journey. This gives young children a clear goal, bite-sized True/False questions, and an intrinsically motivating “advance vs. fall back” loop that reinforces correct safety knowledge.
2. **An under-served hazard domain.** The reviewed systems cover road, fire, and general safety; Toxland VR specifically targets *toxic household materials* (medicines, household poisons, pesticides, aerosols, and supplements) — a leading cause of childhood poisoning that the surveyed literature does not address.
3. **Cross-headset portability.** The same gameplay and educational content, originally built against the Meta SDK for Meta Quest, is re-engineered to run on the PICO XR SDK for the PICO 4 headset. This demonstrates that the educational design is not locked to a single vendor’s hardware ecosystem — the core contribution and the main technical effort of this assignment.

4 Methodology

4.1 Platforms

Toxland VR is developed in **Unity 2022.3.62f3 LTS**. The original source code was a Unity 2019.4.17f1 project targeting the Meta Quest headset through the Meta/Oculus SDK (OVR components). The conversion to PICO 4 required upgrading the Unity editor and replacing the Meta-specific stack with a vendor-neutral XR stack plus the PICO provider. Table 3 lists the platforms and SDKs used.

Table 3: Platforms and SDKs

Component	Version / Detail
Game engine	Unity 2022.3.62f3 LTS (upgraded from 2019.4.17f1)
XR provider	PICO Unity Integration SDK (<code>com.unity.xr.picoxr</code>) via XR Plug-in Management 4.4.0 [5]
Interaction framework	Unity XR Interaction Toolkit 2.5.4 [4]
Input	Unity Input System 1.7.0 (action-based)
UI / text	uGUI + TextMeshPro 3.0.7
Scripting backend	IL2CPP, ARM64
Target platform	Android (PICO 4), minimum SDK level 29 (Android 10)

4.2 Minimum Specification Requirements

Table 4: Minimum system requirements

Requirement	Specification
Target headset	PICO 4
Operating system	Android 10 (API level 29) or higher
Storage	1 GB free space
Input	PICO 4 6DoF controllers (joystick locomotion + ray/crosshair selection)
Graphics	Mobile GPU; anti-aliasing via Unity Quality settings; multi-pass stereo rendering

4.3 System Architecture

Figure 2 shows the high-level architecture. A thin XR input layer (PICO provider → Input System actions → XR Interaction Toolkit interactors) feeds the gameplay managers, which in turn drive the UI and scoring.

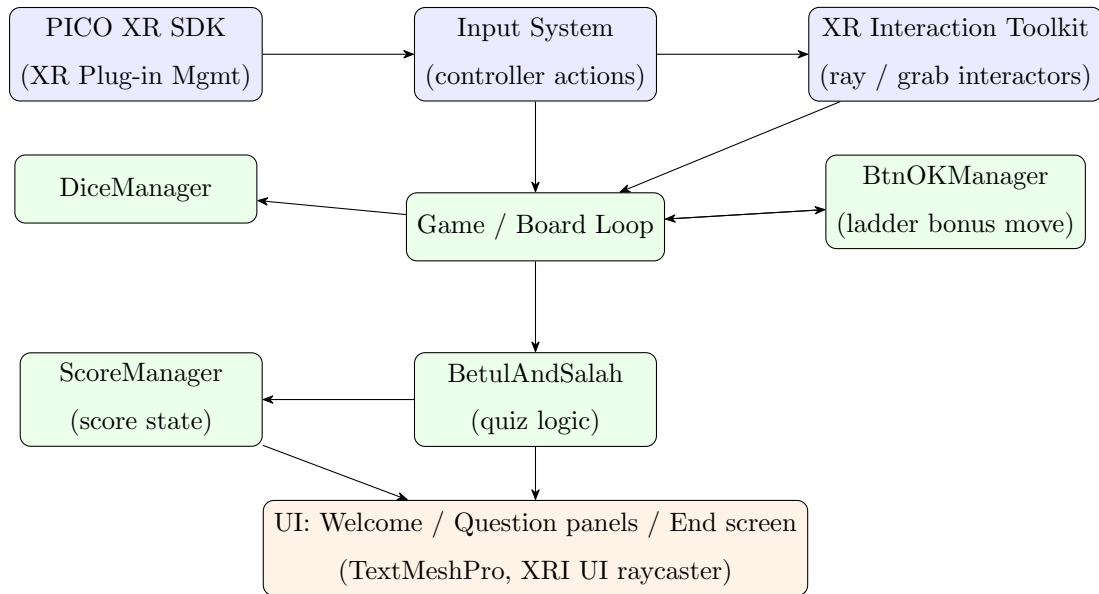


Figure 2: High-level system architecture: XR input layer (blue) → gameplay managers (green) → UI/feedback layer (orange).

4.4 Game Flow

The game flow (Figure 3) is a single loop bounded by the welcome and end screens. On each turn the player rolls the dice and moves along the board to the next space, then resolves that space according to its type. A *Betul/Salah* question space presents a household-safety statement that the player answers *Betul* (true) or *Salah* (false) with the controller ray: a correct answer adds 2 to the score and a wrong answer subtracts 2, after which the score is updated and Right/Wrong feedback with an explanation is shown. Only question spaces change the score; a red box (ladder) presents a good toxicology behaviour and moves the player forward, a blue box (snake) presents an unsafe behaviour and moves the player back (board position only), and a regular space simply passes the turn along. The loop then repeats until the player reaches the final space, where the end screen displays the final score.

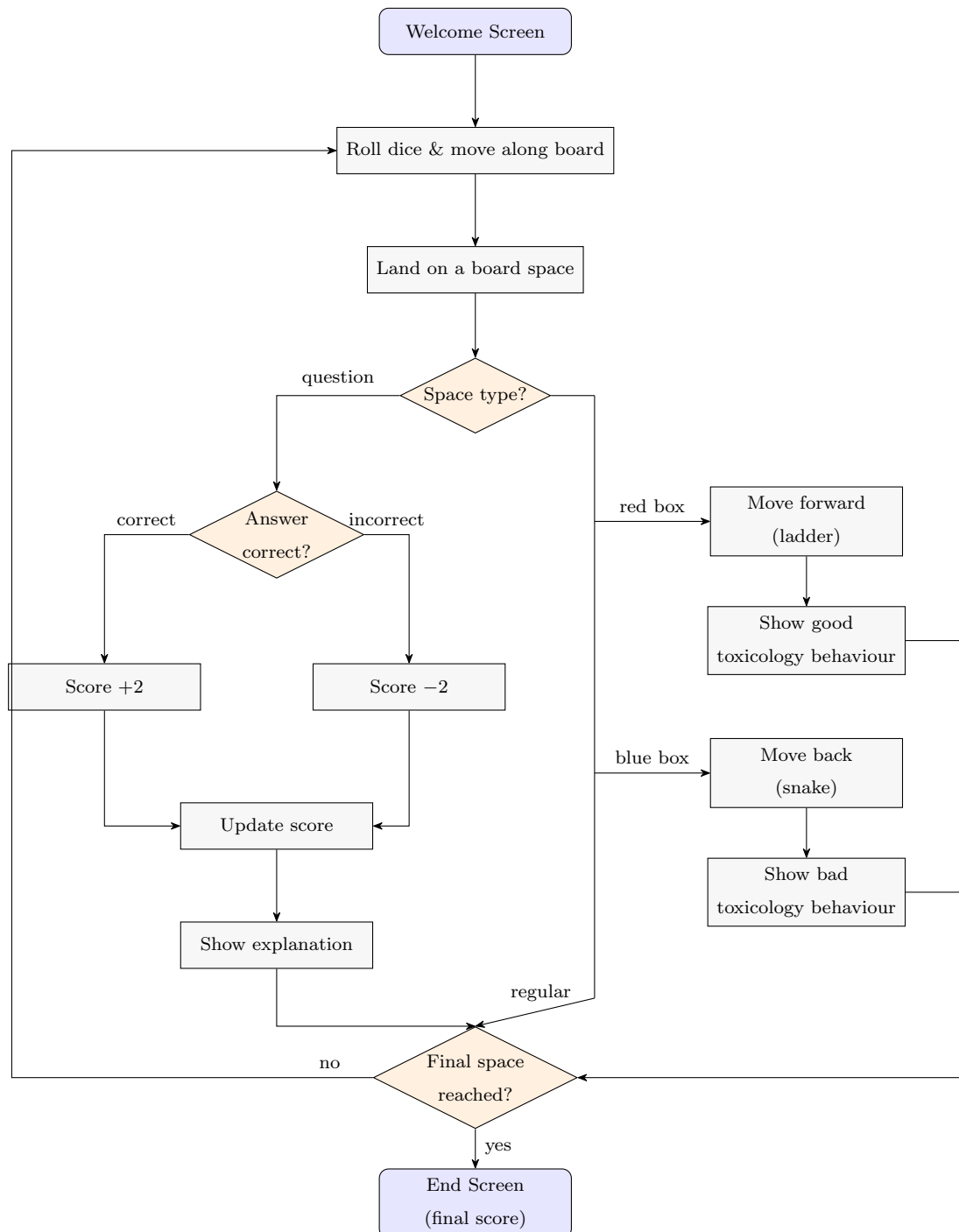


Figure 3: Game flow: each turn the player rolls the dice, moves along the board, and lands on a space that is resolved by type. On a *Betul/Salah* question the player answers and the score changes (+2 if correct, -2 if incorrect) before the score is updated and an explanation is shown. A red box moves the player forward (ladder) and then shows a good toxicology behaviour; a blue box moves the player back (snake) and then shows a bad toxicology behaviour; a regular space passes directly to the final-space check. The loop repeats until the final space is reached.

4.5 Level Design

The level is a **single numbered board**: a winding path of tiles in the Snakes-and-Ladders tradition, running from a Start space to the final space, as illustrated in Figure 4. The tiles are laid out in rows and come in several kinds: plain pass-through spaces, *Betul/Salah* question spaces, red boxes that act as ladders (move forward), and blue boxes that act as snakes (move back), all distributed along the path. The safety content spans the household-hazard themes described above (medicines, household poisons, pesticides and aerosols, supplements, and poisoning response) rather than the board being physically divided into rooms.

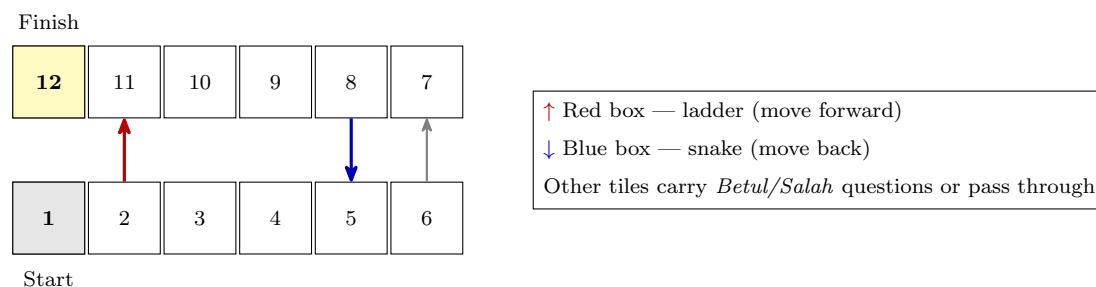


Figure 4: Single numbered Snakes-and-Ladders board (the build lays the tiles out in rows): the player advances from Start to Finish, climbing red-box ladders and sliding down blue-box snakes; other tiles carry *Betul/Salah* household-safety questions or simply pass the turn along.

4.6 Interactive Zone & Triggers

- **Dice trigger** — rolls the dice and produces the movement value (`DiceManager.totalValue`); the player then advances along the board.
- **Red box (ladder) trigger** — detects landing on a red-box tile, presents a good toxicology-related behaviour, and advances the player, granting a bonus move via `BtnOKManager's ExtraMoveEvent`.
- **Blue box (snake) trigger** — detects landing on a blue-box tile, presents a poor or unsafe behaviour, and moves the player back.
- **Question tile trigger** — presents a *Betul/Salah* (True/False) household-safety

statement with Right/Wrong feedback and an explanation panel.

- **Score trigger** — updates `ScoreManager` on correct answers (+2) and wrong answers (−2), and on game completion.

5 System Development

This section describes the implementation and, critically, the **Meta Quest → PICO 4 conversion** performed for each subsystem.

5.1 Conversion Workflow Overview

The migration was carried out in incremental phases, each verified before the next:

1. **Identify and remove Meta dependencies.** The Meta/Oculus SDK and all `OVR` components were removed. A residual `OVRInput` reference in the bundled `IngameDebugConsole` caused a CS0246 compile error and was stripped out.
2. **Install the vendor-neutral XR stack.** XR Plug-in Management 4.4.0, XR Interaction Toolkit 2.5.4, and Input System 1.7.0 were added, and the PICO Unity Integration SDK (`com.unity.xr.picoxr`) was registered as the active XR provider.
3. **Rebuild the camera rig and locomotion** (Section 5.2).
4. **Migrate interaction and UI** to controller-ray input (Section 5.3).
5. **Fix scene flow** — a dead `RoomScene` reference and stale Oculus configuration were removed.
6. **Configure the Android build** — IL2CPP, ARM64, minimum SDK 29, anti-aliasing through Unity Quality settings, PICO platform settings, and the application name (Toxland G6).

5.2 Navigation Control

The Meta `OVRCameraRig` was replaced with a Unity **XR Origin** rig, and the Meta `OVRInput` bindings were replaced with action-based input from the Input System (the

XRI default input actions), with the PICO controller pose actions bound so that head and hand tracking drive the rig. Players move along the board using three complementary mechanisms:

- **Continuous joystick locomotion** via the XRI `DynamicMoveProvider`, driven by the controller thumbstick.
- **Teleportation** to board spaces using XRI Teleport Areas / Teleport Anchors.
- **Dice-roll movement** along the discrete board path.

5.3 Interaction System

VR controller selection is provided by the XRI **XR Ray Interactor**: a pointer ray that the player aims at board spaces, household objects, and UI buttons. During the conversion the ray was made clearly visible to children by widening its line (from 0.005 to 0.02) and setting a cyan valid-target colour. **XR Grab Interactables** allow objects (e.g. the dice) to be picked up. For the educational UI — the *Betul/Salah* quiz panels — an XRI UI input module with a tracked-device graphic raycaster and an `EventSystem` was added so that the controller ray can click world-space `TextMeshPro` buttons; this replaces the Meta interaction framework's UI handling.

5.4 Scoring and Timing System

Scoring is implemented as a lightweight persistent manager. `ScoreManager` holds a static score value and writes it to a `TextMeshPro` display every frame:

```
public class ScoreManager : MonoBehaviour
{
    public static int scoreValue = 0;
    public TextMeshProUGUI score;

    void Update()
    {
        score.text = "Score: " + scoreValue;
    }
}
```

Quiz answers adjust the score through `BetulAndSalah` (+2 for a correct/“Right” answer, −2 for “Wrong”), after which the answer buttons are hidden, a correct/incorrect indicator is shown, and an explanation panel appears after a short delay:

```
public void clickBtn()
{
    btnName.SetActive(false);
    correctWrongName.SetActive(true);
    if (correctWrongName.name == "Right") ScoreManager.scoreValue += 2;
    else if (correctWrongName.name == "Wrong") ScoreManager.scoreValue -= 2;
    Invoke("DisplayExplanation", 3f); // show explanation after 3s
}
```

Ladder bonus moves are delivered through `BtnOKManager`, which invokes a configurable `ExtraMoveEvent` when the player confirms a correct Good-Practice action. The current build uses a **score-based** progression rather than a countdown timer; points accrue from correct safety decisions, quiz answers, hazard identification, and game completion.

5.5 Sound and Feedback

Feedback is delivered through three channels: (i) **audio** — a looping background-music track (`background-music.mp3`) sets a light-hearted tone, and a “Button Pop” click confirms each UI selection; (ii) **visual** — a Right/Wrong indicator image appears immediately after each quiz answer, followed by a written explanation panel; and (iii) **kinaesthetic / spatial** — red-box (ladder) and blue-box (snake) tiles physically move the player forward or backward along the board.

5.6 Welcome Screen and End Screen

The application retains a dedicated welcome screen introducing the game on launch. It is driven by `WelcomeController`, which shows the panel on launch, optionally pauses game time (`Time.timeScale = 0`) while the panel is visible without affecting head tracking or the controller ray, and resumes play when the Start button is clicked:

```
void Awake()
{
```

```
        gameObject.SetActive(true);
        if (pauseWhileVisible) Time.timeScale = 0f;
    }
    public void OnStartClicked()
    {
        if (pauseWhileVisible) Time.timeScale = 1f;
        gameObject.SetActive(false);
    }
}
```

On reaching the final space, the end screen displays the player’s final score. (The online Photon login flow was removed from the build — the build’s scene list contains only **MainScene** — so the application boots directly to the welcome screen; the Photon SDK and login scene remain in the project but are unused in this single-player build.)

5.7 User Manual

1. **Install** the **Toxland.apk** onto the PICO 4 (via SideQuter/ADB `adb install Toxland.apk`, or the PICO Developer tools).
2. **Launch** “Toxland G6” from the PICO library; put on the headset and grip both controllers.
3. On the **Welcome Screen**, aim the controller ray at the **Start** button and pull the trigger.
4. **Move** with the thumbstick (continuous walking) or point the ray at a board space and press teleport. Use the dice to advance along the board.
5. At each space, **read the statement** and answer *Betul* (true) or *Salah* (false) with the controller ray. Watch the Right/Wrong feedback and explanation.
6. Each **correct answer** adds 2 points; each wrong answer subtracts 2.
7. **Reach the final space** to finish; your final score is shown on the end screen.

6 Conclusion

6.1 Challenges and Solutions

The conversion from Meta Quest to PICO 4 surfaced several concrete challenges:

- **Compile-breaking SDK references.** Removing the Oculus SDK left an `OVRInput` reference in the bundled `IngameDebugConsole`, producing a CS0246 error. *Solution:* the Meta-specific input call was removed so the project compiled cleanly against the new XR stack.
- **Editor and toolkit incompatibility.** XR Interaction Toolkit 2.5.4 requires a newer editor than the original Unity 2019.4.17f1. *Solution:* the project was upgraded to Unity 2022.3.62f3 LTS before installing the new packages.
- **Camera rig and input remapping.** The Meta `OVRCameraRig` and `OVRInput` bindings have no direct PICO equivalent. *Solution:* they were rebuilt as an XR Origin rig with action-based Input System bindings and explicitly bound PICO controller pose actions.
- **Controller-driven UI.** The Meta interaction framework previously handled UI; the world-space quiz panels did not respond to the controller ray after migration. *Solution:* an XRI UI input module, tracked-device graphic raycaster, and `EventSystem` were added so the ray can click `TextMeshPro` buttons.
- **Ray visibility for children.** The default interactor ray was too thin to be obvious to young users. *Solution:* the line width was increased and a cyan valid-target colour applied.
- **Scene and build configuration.** A dead `RoomScene` reference and stale Oculus settings broke scene flow, and the Android build needed PICO-appropriate settings. *Solution:* the dead reference was removed and the build configured for IL2CPP/ARM64, minimum SDK 29, mobile-appropriate Unity Quality (anti-aliasing) settings, and PICO platform settings; a test build succeeded and the APK was verified running on a physical PICO 4 (see the recorded presentation).

6.2 Future Improvements

Future work includes: adding voiced narration so pre-literate children can play without reading; expanding the five rooms with more scenarios and hazard variety; introducing adaptive difficulty as in [3]; adding localisation (e.g. Bahasa Melayu / Mandarin); strengthening comfort options (vignetting, snap-turn) to reduce motion sickness in young users; and a formal learning-effectiveness study with a pre/post-test design.

7 VR Prototype

The following screenshots, captured from the running PICO 4 build, demonstrate the functional prototype.

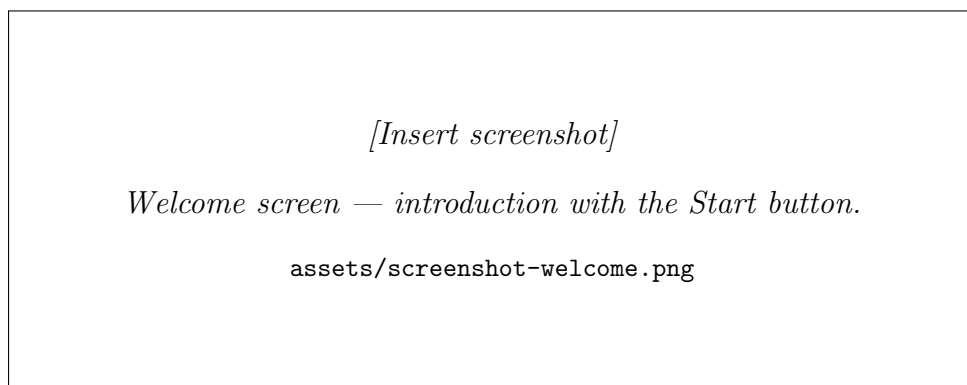


Figure 5: Welcome screen.

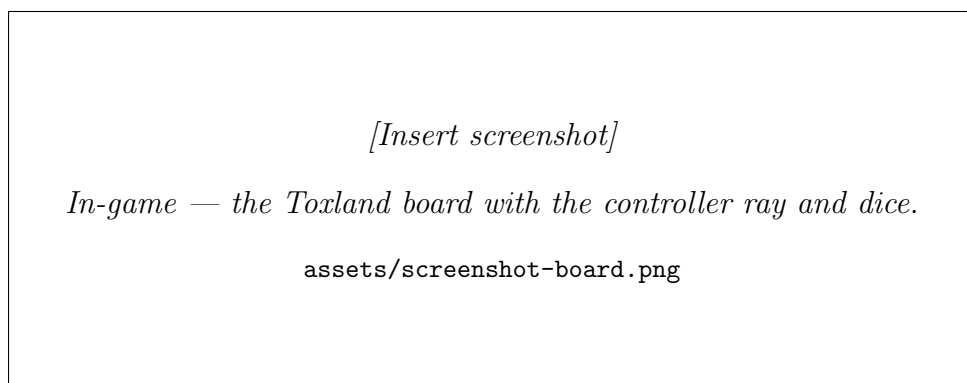


Figure 6: In-game board and navigation.

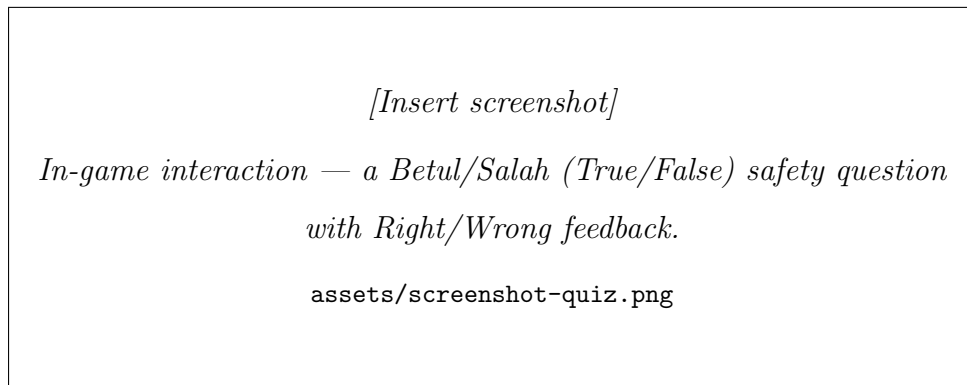


Figure 7: Educational interaction (quiz / safety decision).

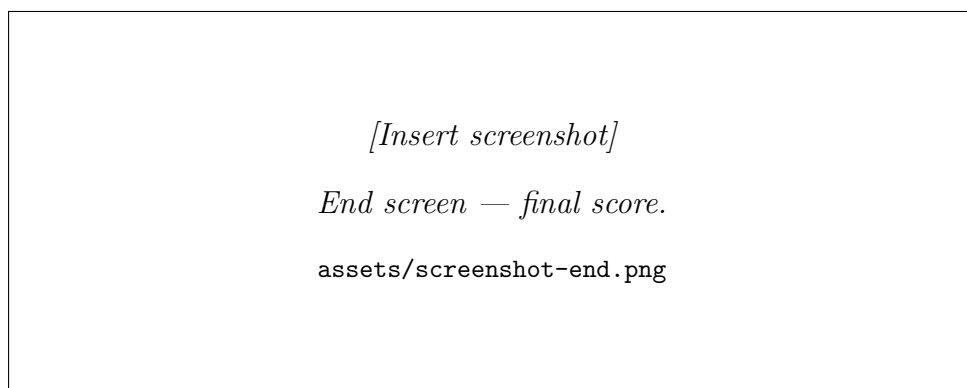


Figure 8: End screen / feedback.

8 Appendix

Table 5: Group members' contributions

Member	Detailed Contributions
Lai Yicheng (163729)	Project Director: project planning, scheduling, task coordination, and final report compilation.
Koay Chun Keat (164592)	Programmer: Unity scripting and game logic; scoring system; Meta-to-PICO migration (XR rig, input, build settings).
Yong Pei Yi, Angeline (163842)	3D Artist: board layout, household-object props, and room environments.
Tan Ke Jun (163204)	Animator: character and object animations (dice, ladders, traps, transitions).
Tan Yi Pin (163382)	UI/UX Designer: welcome/end menus, Betul/Salah question and quiz interfaces, and feedback design.

Members may refine the per-person contribution detail above to match the actual division of work.

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